

Healthcare Claims Analytics | Power BI

An end-to-end Power BI report analyzing **4,500 health insurance claims (\$22.6M billed)** across two fiscal years. It follows claims from submission to decision: where money is denied, which claims are stuck in pending, what conditions drive volume, and how far the underlying data can be trusted.

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Business Questions

The report is designed around the questions a claims or revenue-cycle leader would ask:

1. How much has been billed, and how much of it is **approved, denied, or still pending**?
 2. Where are denials concentrated: by **submission method, procedure type, specialty, and claim type**?
 3. Which **pending claims** have been waiting longest, and how much money is tied up in them?
 4. Which **diagnoses and services** drive claim volume and cost?
 5. Do denial patterns differ across **patient demographics**?
 6. **How reliable is the data** behind all of the above?
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Report Pages

#	Page	What it shows
1	Overview	KPI cards (Total Billed, Approved, Denied, Pending), status mix, fiscal-year trend by quarter, claims map by state, region slicer
2	Denial Analysis	Denial rate by submission method and procedure category, Top 10 denied diagnoses by dollars (Top N filter), specialty × claim type heat map
3	Pending Claims	Pending dollars by aging bucket, pending dollars by specialty and claim type, and a sortable work-queue table of individual pending claims
4	Clinical Mix	Treemap of claims by diagnosis category and diagnosis, most common diagnoses, procedure category by specialty
5	Patient Demographics	Billed amount and denial rate by age band and income band (combo charts), employment status slicer, age × employment heat map
6	Data Quality	Validation-rule results: records affected, dollars affected, % flagged by specialty, and a flagged-records table
7	Claims Details (<i>drill-through</i>)	Detail for a selected diagnosis category or diagnosis: totals, trend by fiscal year, and claim breakdown, with a Back button

Data Model

The source data arrived as a single flat table. I restructured it into a **star schema** with one fact table and supporting dimension tables:

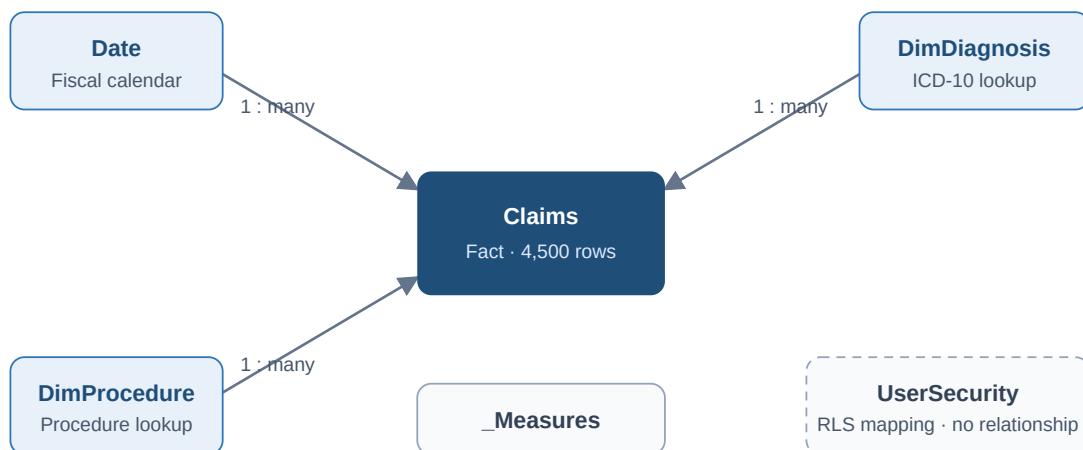


Table	Type	Rows	Purpose
Claims	Fact	4,500	One row per claim: amount, status, dates, provider and patient attributes
Date	Dimension (DAX)	1,096	Fiscal calendar, July 2022 – June 2025
DimDiagnosis	Dimension	4,486	ICD-10 code, official description, plain-language name, category
DimProcedure	Dimension	4,485	Procedure description and category
UserSecurity	Security mapping	7	User email → region, used by dynamic RLS (intentionally not related to other tables)
_Measures	Measure container	0	Holds all 48 measures

All relationships are **one-to-many with single-direction filtering**, from dimension to fact.

Technical Highlights

1. Dedicated Measures Table

All **48 measures** live in a dedicated `_Measures` table instead of being spread across data tables.

- Measures are easy to find: they sit at the top of the Fields pane, grouped by purpose.
- Fact and dimension tables hold only data, which keeps the model clean.
- Naming convention: **columns are always table-qualified** (`Claims[ClaimAmount]`) and **measures never are** (`[Total Billed]`), so the two are easy to tell apart in any formula.
- Measures build on each other instead of repeating logic. For example, `Revenue at Risk = [Denied Amount] + [Pending Amount]` .

```
Total Billed = SUM ( Claims[ClaimAmount] )
Denied Amount = CALCULATE ( [Total Billed], Claims[ClaimStatus] = "Denied" )
Denial Rate = DIVIDE ( [Denied Claims], [Total Claims] )
```

2. Custom Date Table with a July–June Fiscal Year

Auto date/time is turned off. Instead, the model uses a custom DAX date table, **marked as the date table**, with a **fiscal year running July 1 – June 30**.

Why July–June: the claims run from July 2022 to July 2024. A July–June fiscal year gives two complete, comparable years (**FY23 and FY24**). A calendar-year view would show two half years and one full year,

which would make year-over-year comparisons misleading. July–June is also a common fiscal year for hospital systems, academic medical centers, and state governments.

- Built with `GENERATE` + `ROW` so every fiscal attribute comes from one set of variables.
- **Fiscal Year Hierarchy:** Fiscal Year → Fiscal Quarter → Month → Date.
- Sort-by-column on Month (fiscal order, July first), Month Year, Fiscal Year, and Fiscal Year Quarter.
- Time intelligence uses the fiscal year-end, e.g. `TOTALYTD ( [Total Billed], 'Date'[Date], "6/30" )`.

```
VAR fy = IF ( m >= 7, YEAR ( d ) + 1, YEAR ( d ) ) -- July 2023 → FY24
VAR fm = MOD ( m - 7, 12 ) + 1 -- July = fiscal month 1
VAR fq = ROUNDUP ( fm / 3, 0 )
```

3. Row-Level Security (Static and Dynamic)

The model has two approaches to row-level security, to show the trade-off between them.

Static roles, by provider specialty. Each director sees only their own specialty's claims, and a Claims Director role sees everything:

Role	Filter on <code>Claims</code>
Cardiology Director	<code>[ProviderSpecialty] == "Cardiology"</code>
Neurology Director	<code>[ProviderSpecialty] == "Neurology"</code>
Orthopedics Director	<code>[ProviderSpecialty] == "Orthopedics"</code>
Pediatrics Director	<code>[ProviderSpecialty] == "Pediatrics"</code>
General Practice Director	<code>[ProviderSpecialty] == "General Practice"</code>
Claims Director (All Access)	<code>TRUE()</code> (every row passes, so the director sees all claims)

Static roles are simple, but each new specialty means another role to create and maintain. Once a report has security roles, anyone assigned no role sees nothing, so the all-access role makes full access explicit rather than leaving it to chance.

Dynamic role, by region. A single role, `Dynamic Region Access`, serves every user. It looks up the signed-in user's email in the `UserSecurity` mapping table:

```
VAR UserRegions =
    CALCULATETABLE (
        VALUES ( UserSecurity[Region] ),
        UserSecurity[UserEmail] = USERPRINCIPALNAME ()
    )
```

RETURN

```
"All" IN UserRegions  
|| [ProviderRegion] IN UserRegions
```

User (sample)	Region access	Sees
claims.director@contoso.com	All	All regions (Claims Director)
west.manager@contoso.com	West	West only
south.manager@contoso.com	South	South only
midwest.manager@contoso.com	Midwest	Midwest only
northeast.manager@contoso.com	Northeast	Northeast only
coasts.analyst@contoso.com	West, Northeast	Both coasts (multiple rows for one user)

- Granting or changing access means editing a table, not editing the model.
- The "All" value gives the Claims Director full access through the same dynamic role.
- One user can hold several regions.
- UserSecurity has no relationships, so it can't filter or leak into the report visuals.

Testing: Modeling → View as → Other user (enter a sample email) + Dynamic Region Access.

4. Calculated Columns

Ten calculated columns on `Claims` add grouping and slicing attributes. Each is paired with a sort column where needed, so bands display in logical order rather than alphabetically.

Column	Purpose
Age Band / Age Band Sort	0–17, 18–34, 35–49, 50–64, 65+
Income Band / Income Band Sort	Under \$40K through \$130K+
Claim Size Tier	Low / Medium / High claim amount
Is Denied	1/0 flag
Days Pending	Days from claim date to the as-of date (0 if not pending)
Aging Bucket / Aging Bucket Sort	0–30, 31–90, 91–180, 181–365 days, Over 1 year
Data Quality Issue	First validation rule a record fails, or "No issue"

Design choice: calculated columns are used only for values that need to be sliced or grouped (bands, buckets, flags). Anything aggregated is a measure, which isn't stored in the model.

As-of date: aging is calculated against **July 8, 2024**, the last claim date in the data, rather than `TODAY()`. Using today's date would make every claim more than two years old.

5. DAX Patterns

Pattern	Example measures
Filter modification with <code>CALCULATE</code>	Approved / Denied / Pending Amount, Pending Over 1 Year Amount
Safe division with <code>DIVIDE</code>	Denial Rate, Approval Rate, Revenue at Risk %
Industry-standard metric	<code>Adjudicated Denial Rate</code> excludes pending claims from the denominator (Denied ÷ (Approved + Denied)), alongside the simple Denial Rate
Fiscal time intelligence	Billed FYTD, Billed PY, Billed YoY %, Denial Rate YoY Change (pts)
Rolling window	<code>Billed Rolling 3M Avg</code> using <code>DATESINPERIOD</code> + <code>AVERAGEX</code>
Share of total	<code>% of Total Billed</code> using <code>ALLSELECTED</code>
Ranking	<code>Specialty Denial Rank</code> and <code>State Revenue at Risk Rank</code> using <code>RANKX</code> with <code>HASONEVALUE</code> guards
Top-N lookup	<code>Top Diagnosis</code> and <code>Top Procedure</code> using <code>TOPN</code> + <code>CONCATENATEX</code>
Dynamic text	Selected region and fiscal-year labels, page titles, and narrative "insight" sentences that update with slicers

The full DAX for every measure, column, the date table, and the security roles is in [DAX_Reference.md](#).

6. Data Quality Validation

Before trusting the analysis, I tested the data against logical validation rules and gave the results their own page:

Rule	Records
Adult (18+) treated under Pediatrics	787
Married / Widowed / Divorced under age 16	532
"Retired" under age 50	520
"Employed" under age 14	144
Records failing at least one rule	1,673 (37.2%)
Dollars affected	\$8.4M

- The rule counts add up to 1,983, more than the 1,673 records flagged, because **310 records break more than one rule**. Each rule is counted separately, and a separate measure counts unique flagged records.
- **95.7% of Pediatrics claims are for adult patients**, so specialty-level conclusions about Pediatrics in this dataset should not be relied on.
- Findings that involve flagged fields are called out as caveats. For example, the high denial rate for "retired" patients aged 18–34 on the Demographics page comes from records that fail validation.
- A flagged-records table lists each affected claim and its issue, ready to be routed back to the source system for correction.

7. Navigation and Interactivity

- **Page navigation buttons** on every page, so the report works like an app.
- **Drill-through** from Clinical Mix to a Claims Details page by diagnosis category or diagnosis, with a **Back** button.
- **Top N filtering**: the Top 10 denied diagnoses update with every slicer selection.
- **Conditional formatting heat maps**: specialty × claim type and age band × employment status, colored by denial rate.
- **Combo charts** showing dollars (columns) and denial rate (line, secondary axis) together.
- **Slicers** for provider region and patient employment status.
- A **custom report theme** applied across all pages.

8. Performance

Tested with Performance Analyzer:

- **DAX query time**: ~10 ms per visual.
 - Visuals render in **under 150 ms** once cached.
 - Most of the first-load time is Power BI startup overhead after a model change, not measure calculation.
 - Heavy patterns (`RANKX` , `TOPN`) are kept off large tables and matrices, where they would recalculate for every row.
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Key Findings

Metric	Value
Total billed	\$22.6M (4,500 claims)
Approved	\$7.8M (35%)
Denied	\$7.6M (33%)
Pending	\$7.2M (32%)
Revenue at risk (denied + pending)	\$14.8M (65%)

- **Submission method:** Online submissions have the highest denial rate (35.9%), Phone the lowest (31.8%).
- **Procedure type:** Diagnostic Testing has the highest denial rate (36.8%); Therapeutic and Surgical/Interventional the lowest (~30%).
- **Specialty × claim type:** General Practice emergency claims have the highest denial rate of any combination (40.4%). Orthopedics is consistently lowest (~30%) across all claim types.
- **Pending backlog:** 743 of 1,466 pending claims (\$3.6M) have been waiting **more than a year**. Average time pending is ~368 days.
- **Most common diagnoses:** high blood pressure (293 claims), common cold/upper respiratory infection (233), and lower back pain (186).
- **Top denied diagnosis by dollars:** high blood pressure (\$513K denied).

Note: The base dataset is synthetic, and its categories are close to evenly distributed. The differences above are modest, so they are presented as patterns the report surfaces rather than causal conclusions. The project's focus is the model, the DAX, the security, and the analytical approach.

Data Sources and Preparation

Base dataset: a publicly available synthetic health insurance claims dataset (4,500 rows, 17 columns, July 2022 – July 2024). It contains no real patient information.

Enrichment and preparation:

Step	Why
Replaced 3,876 auto-generated placeholder city names with real U.S. cities and states (65 cities, 42 states + DC), weighted by population so cities repeat realistically	Enables state-level mapping and analysis
Added a ProviderRegion column (Northeast, Midwest, South, West)	Natural dimension for regional reporting and dynamic RLS
Built DimDiagnosis mapping each diagnosis code to a real ICD-10-CM code , official description, plain-language name, and ICD-10 chapter category, matched to the claim's specialty	Original codes were random strings with no meaning
Built DimProcedure with procedure descriptions and categories (CPT codes intentionally not used, as they are licensed by the AMA)	Enables procedure-level analysis
De-duplicated codes that differed only by letter case	Power BI relationships are case-insensitive, so these would otherwise cause duplicate-key errors
Created UserSecurity mapping table with sample users	Drives dynamic RLS

Assumptions:

- Fiscal year runs **July 1 – June 30** (FY24 = July 2023 – June 2024).
- Aging is calculated as of **July 8, 2024**, the last date in the data.
- Diagnosis and procedure descriptions are **synthetic assignments** for analysis purposes. They are not the original meaning of the codes.

How to Open the Report

1. Install **Power BI Desktop** (free, Windows).
2. Download this repository and open **ClaimsProject.pbix**.
3. The report opens with data already loaded. To refresh from the source files, point the report to your copy of the **data/** folder: **Transform data** → **Data source settings** → **Change Source**.
4. To test security: **Modeling** → **View as**, then choose a specialty role, or choose **Other user** with a sample email from **UserSecurity** plus **Dynamic Region Access**.

The Overview map uses the **ArcGIS for Power BI** visual, which may ask you to accept its terms the first time it loads.

Repository Structure

```
├─ ClaimsProject.pbix          # Power BI report
├─ README.md                  # This file
├─ DAX_Reference.md           # All DAX: measures, calculated columns, date table, RLS
├─ data/
│   ├─ health_insurance_claims_us_cities.xlsx
│   ├─ DimDiagnosis.xlsx
│   ├─ DimProcedure.xlsx
│   └─ UserSecurity_roles.xlsx
└─ images/                    # Report and model screenshots
```

Skills Demonstrated

Data modeling: star schema design · relationship cardinality and filter direction · custom date table · calculated tables

DAX: 48 measures covering filter context, fiscal time intelligence, ranking, rolling averages, dynamic text, and validation logic

Security: static and dynamic row-level security with `USERPRINCIPALNAME()` and a mapping table

Data preparation: Power Query · data enrichment · dimension table design · key de-duplication

Data quality: validation-rule design · impact quantification · documented caveats

Report design: page navigation · drill-through · Top N filtering · conditional formatting · combo charts · custom theme · performance testing

Questions or feedback are welcome.